



HUST

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ONE LOVE. ONE FUTURE.



FUNDAMENTALS OF OPTIMIZATION



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Convex Optimization

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Outline

- Basis
- Unconstrained Optimization
- Constrained Optimization



Notations

- $x \in R^n$: vector of decision variables $x_j, j = 1, 2, \dots, n$
- $f: R^n \rightarrow R$ is the objective function (**dom** $f = R^n$)
- $g_i: R^n \rightarrow R$ is the constraint function defining restrictions on $x, i = 1, 2, \dots, m$

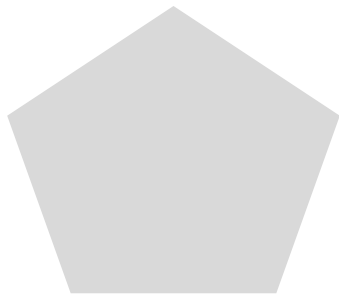
minimize $f(x)$ over $x = (x_1, x_2, \dots, x_n) \in X \subset R^n$ satisfying a property P :

$$g_i(x) \leq b_i, i = 1, 2, \dots, s$$

$$g_i(x) = d_i, i = s + 1, 2, \dots, m$$

Notations

- S is called a convex set if: $\forall u_1, u_2, \dots, u_k$ in S , $\forall$ non-negative numbers $\lambda_1, \lambda_2, \dots, \lambda_k$ such that $\sum_{i=1}^k \lambda_i = 1$, then $\sum_{i=1}^k u_i \lambda_i$ is in S



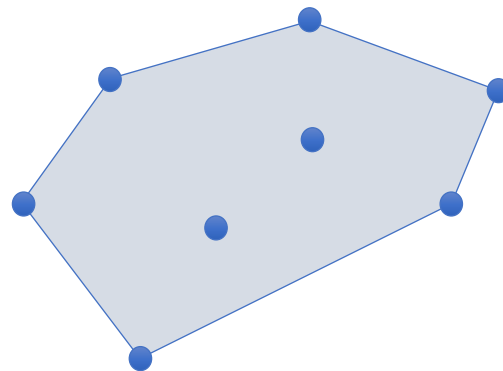
Convex set



Non Convex set

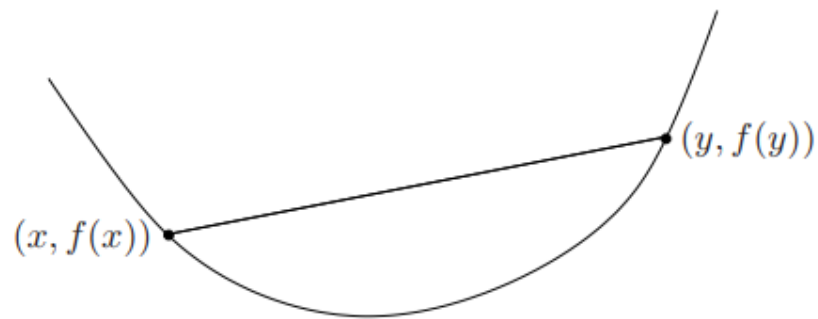
Notations

- We say that $y \in \mathbb{R}^n$ is a **convex combination** of the points $x^{(1)}, \dots, x^{(k)} \in \mathbb{R}^n$ if $y = \lambda_1 x^{(1)} + \dots + \lambda_k x^{(k)}$ for some choice of $\lambda_1, \dots, \lambda_k \in [0,1]$ such that $\lambda_1 + \dots + \lambda_k = 1$
- Let C be the set of all points y that can be obtained as a convex combination of the $x^{(1)}, \dots, x^{(k)}$
 - In the special case $k = 2$, C is just a line segment
 - C is a convex set called the **convex hull** of the $x^{(1)}, \dots, x^{(k)}$



Notations

- Linear function: $f(x) = Ax$
- Affine function: $f(x) = Ax + b$
- Convex function
 - f is called convex if $\forall x_1, x_2$ and $\forall \lambda \in (0, 1)$:
$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2)$$
 - f is called strictly convex if $\forall x_1 \neq x_2$ and $\forall \lambda \in (0, 1)$:
$$f(\lambda x_1 + (1 - \lambda)x_2) < \lambda f(x_1) + (1 - \lambda)f(x_2)$$



Notations

- Example: $f(x) = 2x + 3$
$$\begin{aligned}f(\lambda x_1 + (1 - \lambda)x_2) &= 2(\lambda x_1 + (1 - \lambda)x_2) + 3 \\&= (2\lambda x_1 + 3\lambda) + (2(1 - \lambda)x_2) + (1 - \lambda)3 \\&= \lambda f(x_1) + (1 - \lambda)f(x_2)\end{aligned}$$

Notations

- Examples
 - $f(x) = x^2$
 - $f(x) = e^{ax}$, a is a constant
 - $f(x) = x \ln x$

Notations

- $f(x_1, x_2, \dots, x_n)$ is a multivariable function

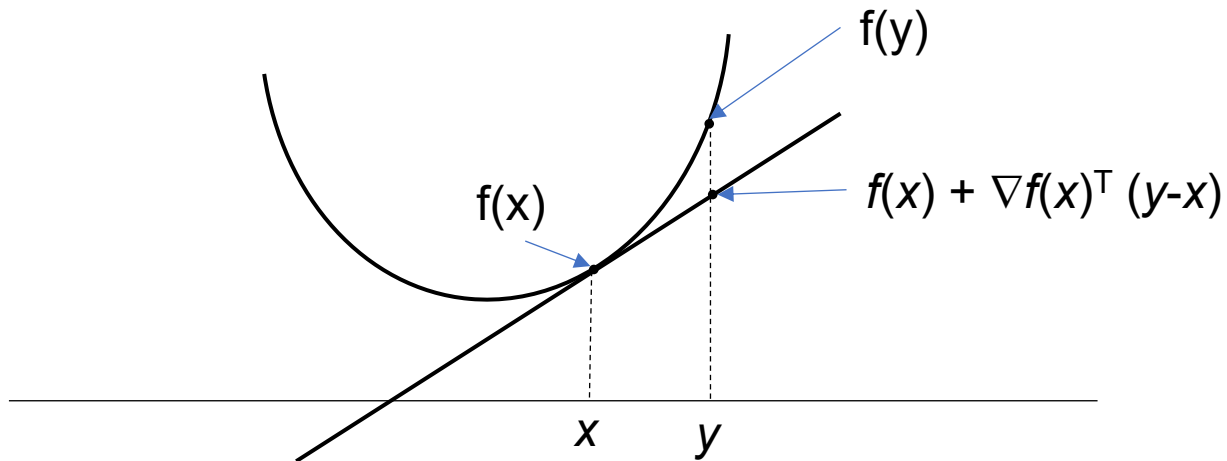
$$\nabla f(x) \text{ (or } f'(x)) = \begin{pmatrix} \frac{\partial f}{\partial x_1}(x) \\ \frac{\partial f}{\partial x_2}(x) \\ \dots \\ \frac{\partial f}{\partial x_n}(x) \end{pmatrix}$$

$$\nabla^2 f(x) \text{ (or } f''(x)) = \frac{\partial^2 f}{\partial x_i \partial x_j}(x) \text{ called Hessian matrix}$$

Notations

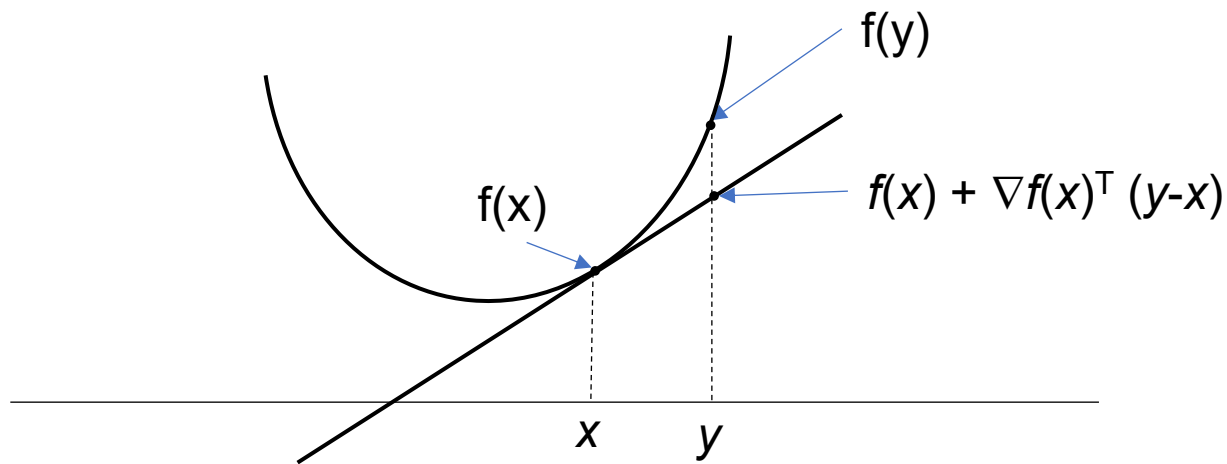
- If f is convex and differentiable, then

$$f(x) + \nabla f(x)^\top (y-x) \leq f(y), \forall x, y \in \text{dom } f$$



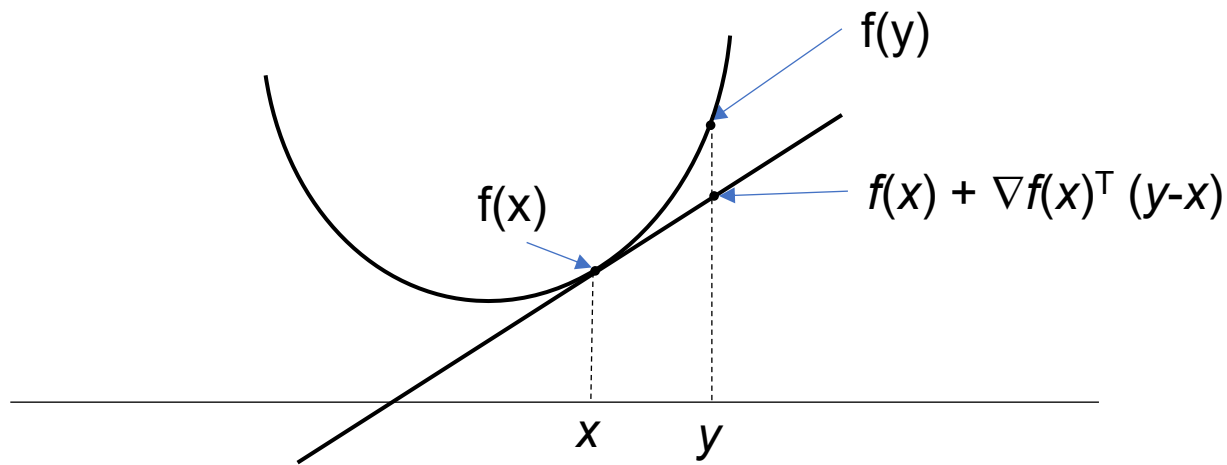
Notations

- First-order condition
 - Suppose f is differentiable (i.e., its gradient exists at all points in $\text{dom } f$, which is open). f is convex if and only if $\text{dom } f$ is convex and $f(x) + \nabla f(x)^\top (y-x) \leq f(y)$, $\forall x, y \in \text{dom } f$



Notations

- First-order condition
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Notations

- Proof (First-order condition, case $n = 1$)

- Suppose f is differentiable and is convex

- $\forall 0 < t \leq 1, x + t(y-x) \in \text{dom } f$

- By convexity of f , $\forall x, y \in \text{dom } f$, we have

$$f(x + t(y-x)) \leq (1-t) f(x) + t f(y)$$

$$\rightarrow f(y) \geq f(x) + \frac{f(x+t(y-x))-f(x)}{t} = f(x) + \frac{f(x+t(y-x))-f(x)}{t(y-x)}(y-x). \text{ By taking the limit as } t(y-x) \rightarrow 0 \text{ yields } f(x) + \nabla f(x)^T (y-x) \leq f(y)$$

- Suppose $f(x) + \nabla f(x)^T (y-x) \leq f(y)$, $\forall x, y \in \text{dom } f$

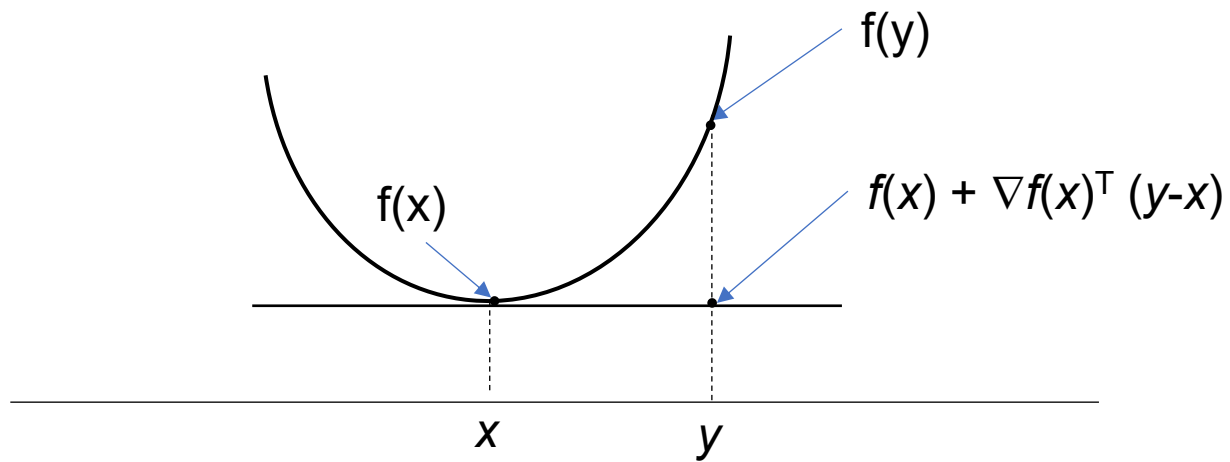
- Let $z = \lambda x + (1-\lambda)y$

- $f(x) \geq f(z) + f'(z) (x-z)$, $f(y) \geq f(z) + f'(z) (y-z) \rightarrow \lambda f(x) + (1-\lambda) f(y) \geq f(z) = f(\lambda x + (1-\lambda)y) \rightarrow f$ is convex

- Proof (First-order condition, general case)
 - Given $x, y \in \mathbb{R}^n$, denote $g(t) = f(ty + (1-t)x)$, $g'(t) = \nabla f(ty + (1-t)x)(y-x)$
 - (1) Assume f is convex $\rightarrow g$ is convex. Hence $g(1) \geq g(0) + g'(0)$ (see case $n = 1$ above). This implies $f(y) \geq f(x) + \nabla f(x)^T(y-x)$
 - (2) Assume $f(y) \geq f(x) + \nabla f(x)^T(y-x)$, $\forall x, y \in \text{dom } f$
 - If $py + (1-p)x \in \text{dom } f$ and $qy + (1-q)x \in \text{dom } f$, we have
$$f(py + (1-p)x) \geq f(qy + (1-q)x) + \nabla f(qy + (1-q)x)^T(y-x)(p-q)$$
i.e., $g(p) \geq g(q) + g'(q)(p-q) \rightarrow g$ is convex (this implies f is convex)

Notations

- If $\nabla f(x) = 0$, then $f(y) \geq f(x)$, $\forall y \in \text{dom } f \rightarrow x$ is global minimizer of the function f



Norms

- Norm: A real-valued function $f(x)$ on R^n is called a norm, if
 - $f(x) \geq 0$
 - $\lambda f(x) = f(\lambda x)$
 - $f(x + y) \leq f(x) + f(y)$ (triangle inequality)
- Examples
 - $\|x\|_p = (|x_1|^p + |x_2|^p + \dots + |x_n|^p)^{1/p}$
 - $\|x\|_1 = (|x_1| + |x_2| + \dots + |x_n|)$
 - $\|x\|_\infty = \max\{|x_1|, |x_2|, \dots, |x_n|\}$

Taylor approximation

- Single variable Taylor series

$$f(x) = f(a) + \frac{f'(a)}{1!} (x - a) + \frac{f''(a)}{2!} (x - a)^2 + \dots$$

- First-order Taylor approximation

$$f(x + h) \approx f(x) + h^T \nabla f(x)$$

- Second-order Taylor approximation

$$f(x+h) \approx f(x) + h^T \nabla f(x) + \frac{1}{2} h^T \nabla^2 f(x) h$$

Unconstrained convex optimization

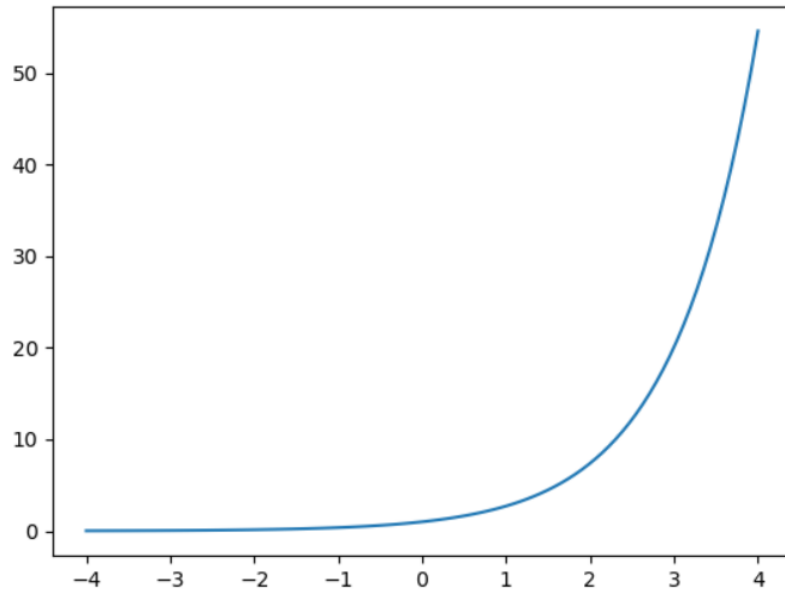
- Unconstrained, smooth convex optimization problem:

$$\min f(x)$$

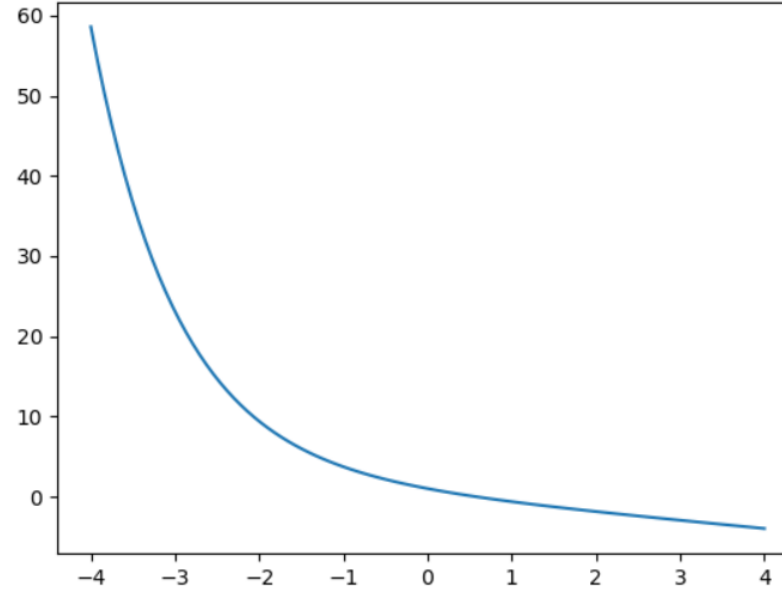
- $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is convex and twice differentiable
- dom** $f = \mathbb{R}^n$: no constraint
- Assumption: the problem is solvable with $f^* = \min_x f(x)$ and $x^* = \arg\min_x f(x)$
- To find x , solve equation $\nabla f(x^*) = 0$: not easy to solve analytically
- Iterative scheme is preferred: compute minimizing sequence $x^{(0)}, x^{(1)}, \dots$ s.t. $f(x^{(k)}) \rightarrow f(x^*)$ as $k \rightarrow \infty$
- The algorithm stops at some point $x(k)$ when the error is within acceptable tolerance: $f(x^{(k)}) - f^* \leq \varepsilon$

Local minimizer

- x^* is a local minimizer for $f: R^n \rightarrow R$ if $f(x^*) \leq f(x)$ for $\|x^* - x\| \leq \varepsilon$ ($\varepsilon > 0$ is a constant)
- x^* is a global minimizer for $f: R^n \rightarrow R$ if $f(x^*) \leq f(x)$ for all $x \in R^n$



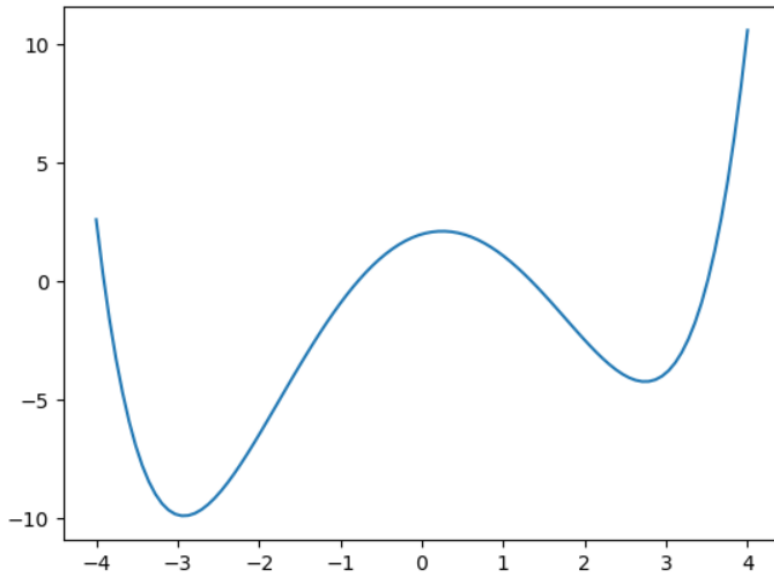
$f(x) = e^x$ has no minimizer



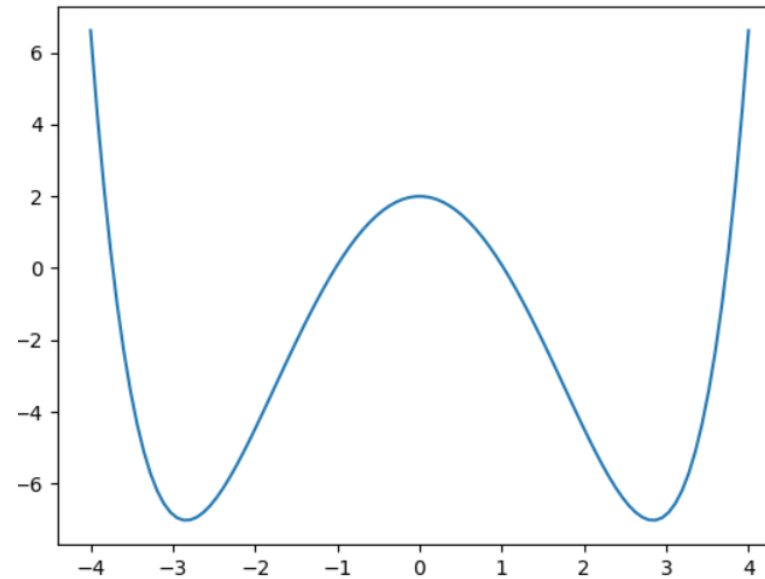
$f(x) = -x + e^{-x}$ has no minimizer

Local minimizer

- x^* is a local minimizer for $f: R^n \rightarrow R$ if $f(x^*) \leq f(x)$ for $\|x^* - x\| \leq \varepsilon$ ($\varepsilon > 0$ is a constant)
- x^* is a global minimizer for $f: R^n \rightarrow R$ if $f(x^*) \leq f(x)$ for all $x \in R^n$



$f(x) = e^x + e^{-x} - 3x^2 + x$ has one local minimizer and one global minimizer



$f(x) = e^x + e^{-x} - 3x^2$ has two global minimizers

Local minimizer

- **Theorem** (Necessary condition for local minimum) If x^* is a local minimizer for $f: R^n \rightarrow R$, then $\nabla f(x^*) = 0$ (x^* is also called *stationary point* for f)
- **Proof** Suppose that x^* is a local minimizer but $\nabla f(x^*) \neq 0$.
 - We can find a vector z such that $\langle \nabla f(x^*), z \rangle < 0$ (for instance $z = -\nabla f(x^*)$, $\langle \nabla f(x^*), z \rangle = -\|\nabla f(x^*)\|^2 < 0$)
 - For a constant $t \geq 0$, consider vector $x(t) = x^* + tz$, and $\varphi(t) = f(x(t))$
 - $\frac{d\varphi(t)}{dt} \Big|_{t=0} = \frac{df(x(t))}{dt} \Big|_{t=0} = \langle \nabla f(x^*), z \rangle < 0 \rightarrow$ with constant t small enough, $\varphi(t) < \varphi(0)$ or $f(x(t)) < f(x^*)$ (conflict with the assumption that x^* is a local minimizer) $\square$

Local minimizer

Example

- $f(x,y) = x^2 + y^2 - 2xy + x$
- $\nabla f(x) = \begin{pmatrix} 2x - 2y + 1 \\ 2y - 2x \end{pmatrix} = 0$ has no solution

→ there is no minimizer of $f(x,y)$

Example

- $f(x,y) = x^2 + y^2 - 2xy + x$
- $\nabla f(x) = \begin{pmatrix} 2x - 2y + 1 \\ 2y - 2x \end{pmatrix} = 0$ has no solution

→ there is no minimizer of $f(x,y)$

Local minimizer

- **Theorem** (Sufficient condition for a local minimum) Assume x^* is a stationary point and that $\nabla^2 f(x^*)$ is positive definite, then x^* is a local minimizer

$$\nabla^2 f(x) = \begin{pmatrix} \frac{\partial^2 f(x)}{\partial x_1 \partial x_1} & \frac{\partial^2 f(x)}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f(x)}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f(x)}{\partial x_2 \partial x_1} & \frac{\partial^2 f(x)}{\partial x_2 \partial x_2} & \cdots & \frac{\partial^2 f(x)}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f(x)}{\partial x_n \partial x_1} & \frac{\partial^2 f(x)}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f(x)}{\partial x_n \partial x_n} \end{pmatrix}$$

Local minimizer

- Matrix $A_{n \times n}$ is called positive definite if

$$A^i = \begin{pmatrix} a_{1,1} & a_{1,2} & \dots & a_{1,i} \\ a_{2,1} & a_{2,2} & \dots & a_{2,i} \\ \dots & \dots & \dots & \dots \\ a_{i,1} & a_{i,2} & \dots & a_{i,i} \end{pmatrix}, \det(A^i) > 0, i = 1, \dots, n$$

Local minimizer

- **Example** $f(x,y) = e^{x^2+y^2}$

$$\nabla f(x,y) = \begin{pmatrix} 2xe^{x^2+y^2} \\ 2ye^{x^2+y^2} \end{pmatrix} = 0 \text{ has unique solution } x^* = (0,0)$$

$$\nabla^2 f(x,y) = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} > 0 \rightarrow (0,0) \text{ is a minimizer of } f$$

Local minimizer

- **Example** $f(x,y) = x^2 - y^2 + 2xy - x$

$$\nabla f(x,y) = \begin{pmatrix} 2x + 2y - 1 \\ 2x - 2y \end{pmatrix} = 0$$

has unique solution $x^* = (1/4, 1/4)$

$$\nabla^2 f(x^*, y^*) = \begin{pmatrix} 2 & 2 \\ 2 & -2 \end{pmatrix} \quad \text{is not positive definite}$$

→ cannot conclude x^*

Descent method

Determine starting point $x^{(0)} \in R^n$;

$k \leftarrow 0$;

while(stop condition not reach){

 Determine a search direction $p_k \in R^n$;

 Determine a step size $\alpha_k > 0$ s.t. $f(x^{(k)} + \alpha_k p_k) < f(x^{(k)})$;

$x^{(k+1)} \leftarrow x^{(k)} + \alpha_k p_k$;

$k \leftarrow k+1$;

}

Stop condition may be

- $\|\nabla f(x^k)\| \leq \varepsilon$
- $\|x^{k+1} - x^k\| \leq \varepsilon$
- $k > K$ (maximum number of iterations)

Gradient descent method

- Gradient descent schema

$$\mathbf{x}^{(k)} = \mathbf{x}^{(k-1)} - \alpha_k \nabla f(\mathbf{x}^{(k-1)})$$

```
init  $\mathbf{x}^{(0)}$ ;  
 $k = 1$ ;  
while stop condition not reach{  
    specify constant  $\alpha_k$ ;  
     $\mathbf{x}^{(k)} = \mathbf{x}^{(k-1)} - \alpha_k \nabla f(\mathbf{x}^{(k-1)})$ ;  
     $k = k + 1$ ;  
}
```

- α_k might be specified in such a way that $f(\mathbf{x}^{(k-1)} - \alpha_k \nabla f(\mathbf{x}^{(k-1)}))$ is minimized: $\frac{\partial f}{\partial \alpha_k} = 0$

Gradient descent method

- $f(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2 - x_1x_2 - x_2x_3 + x_1 + x_3$
- $\nabla f(x_1, x_2, x_3) = [2x_1 - x_2 + 1, 2x_2 - x_1 - x_3, 2x_3 - x_2 + 1]^T$

- $\nabla^2 f = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$

Gradient descent method

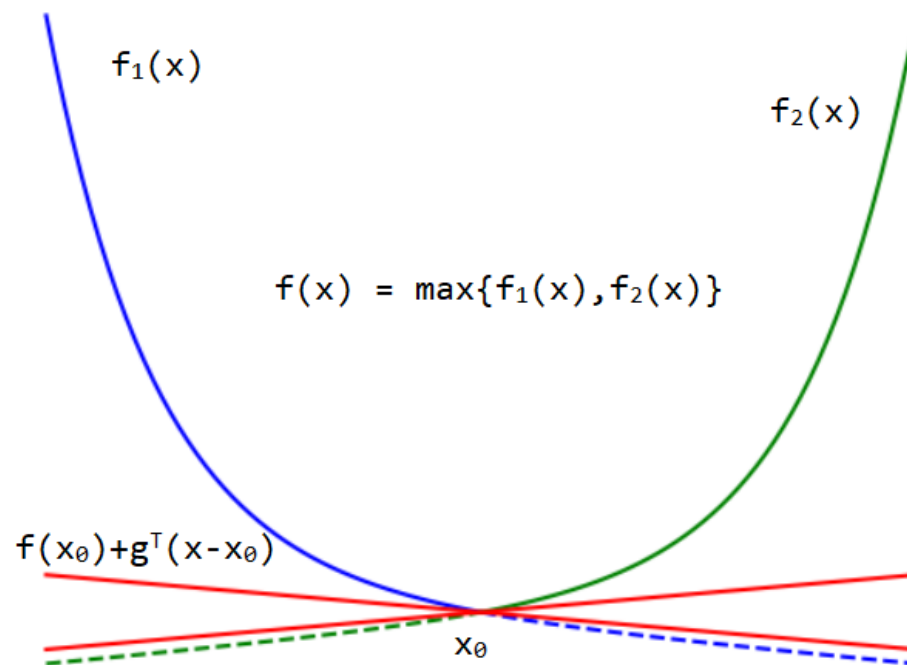
```
import numpy as np
f = lambda x1,x2,x3: x1**2 + x2**2 + x3**2 - x1*x2 - x2*x3 + x1 + x3 # function f
df = lambda x1,x2,x3: [2*x1 + 1 - x2, -x1 + 2*x2 - x3, -x2 + 2*x3 + 1] # gradient
x1,x2,x3 = 0,0,0
for i in range(1000):
    [D1,D2,D3] = df(x1,x2,x3)
    A = 2*x1*D1 + 2*x2*D2 + 2*x3*D3 - x1*D2 - x2*D1 - x2*D3 -x3*D2 + D1 + D3
    B = 2*D1*D1 + 2*D2*D2 + 2*D3*D3 -2*D1*D2 - 2*D2*D3
    if B == 0:
        break
    alpha = A/B
    x1 = x1 - alpha*D1
    x2 = x2 - alpha*D2
    x3 = x3 - alpha*D3
```

Subgradient method

- For minimize nondifferentiable convex function
- Subgradient method is not a descent method: the function value can increase

Subgradient method

- Subgradient of f at x
 - Any vector g such that $f(x') \geq f(x) + g^T(x'-x)$
 - If f is differentiable, only possible choice is $g^{(k)} = \nabla f(x^{(k)})$, $\rightarrow$ the subgradient method reduces to the gradient method



Basic Subgradient method

$$x^{(k+1)} = x^{(k)} - \alpha_k g^{(k)}$$

- $x^{(k)}$: is at the k^{th} iteration
- $g^{(k)}$: any subgradient of f at $x^{(k)}$
- $\alpha_k > 0$ is the k^{th} step size
- Note: subgradient is not a descent method, thus $f_{best}^{(k)} = \min\{f(x^{(1)}), f(x^{(2)}), \dots, f(x^{(k)})\}$

Basic Subgradient method - example

$$\text{minimize } f(x) = \max_{i=1,\dots,m}(a_i^T x + b_i)$$

- Finding subgradient: given x , the index j for which

$$a_j^T x + b_j = \max_{i=1,\dots,m}(a_i^T x + b_i)$$

→ subgradient at x is $g = a_j$

Basic Subgradient method - example

```
import numpy as np

def solve(A,b):
    f = lambda x: np.max(A.dot(x) + b)
    sg = lambda x: A[np.argmax(A.dot(x) + b)]
    x0 = [0,0,0,0]
    x = np.array(x0).T
    f_best = f(x)
    for i in range(100000):
        alpha = 2
        x = x - alpha*sg(x)
        if f_best > f(x):
            f_best = f(x)
    return f_best
```

Basic Subgradient method - example

```
def main():  
    A = np.array([[1,-2,3,-5],[2,-2,1,1],[-3,2,-2,7]],dtype='double')  
    b = np.array([3,4,5]).T  
    rs = solve(A,b)  
    print('rs = ',rs)  
  
if __name__ == '__main__':  
    main()
```

Convex constrained optimization

- Lagrange dual function
- KKT condition



Lagrange dual function

- Optimization problem in the standard form

$$\begin{array}{ll} (P) & \text{minimize } f(x) \\ \text{s.t.} & g_i(x) \leq 0, i = 1, 2, \dots, m \\ & h_i(x) = 0, i = 1, 2, \dots, p \\ & x \in X \subseteq R^n \end{array}$$

with $x \in R^n$, and assume $D = (\cap_{i=1}^m \text{dom } g_i) \cap (\cap_{i=1}^m \text{dom } h_i)$ is not empty.

- Denote f^* the optimal value of $f(x)$
- If $f, g_i (i = 1, 2, \dots, m)$ are convex functions, $h_i (i = 1, \dots, p)$ are linear $\rightarrow$ **convex program**

Lagrange dual function

- Define Lagrangian function $L: \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p \rightarrow \mathbb{R}$

$$L(x, \lambda, \mu) = f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{i=1}^p \mu_i h_i(x)$$

- Lagrange dual function (or dual function)

$$q(\lambda, \mu) = \inf_{x \in D} L(x, \lambda, \mu)$$

Lagrange dual function

- **Weak duality theorem** if x^* is an optimal solution to the primal problem and (λ^*, μ^*) is an optimal solution to the dual problem, then $f(x^*) \geq q(\lambda^*, \mu^*)$
- **Corollary** If there exist x^* and (λ^*, μ^*) such that $f(x^*) = q(\lambda^*, \mu^*)$, then x^* and (λ^*, μ^*) are respectively optimal solutions to the primal and dual problems

KKT conditions

- **Theorem** (Fritz John necessary conditions) Let x^* be a feasible solution of (P) . If x^* is a local minimum of (P) , then there exists (u, λ, μ) such that:
 - $u \nabla f(x^*) + \sum_{i=1}^m \lambda_i \nabla g_i(x^*) + \sum_{i=1}^p \mu_i \nabla h_i(x^*) = 0$
 - $u, \lambda \geq 0, (u, \lambda, \mu) \neq 0$
 - $\lambda_i g_i(x^*) = 0, i = 1, \dots, m$

KKT conditions

- **Theorem** (Karush-Kuhn-Tucker (KKT) necessary conditions) Let x^* be a feasible solution of (P) and $I = \{i: g_i(x^*) = 0\}$. Further, suppose that $\nabla h_i(x^*)$ for $i = 1, \dots, p$ and $g_i(x^*)$ for $i \in I$ are linearly independent. If x^* is a local minimum of (P) , then there exists (λ, μ) such that:
 - $\nabla f(x^*) + \sum_{i=1}^m \lambda_i \nabla g_i(x^*) + \sum_{i=1}^p \mu_i \nabla h_i(x^*) = 0$
 - $\lambda \geq 0$,
 - $\lambda_i g_i(x^*) = 0, i = 1, \dots, m$

KKT conditions

- **Theorem** (KKT sufficient conditions) Let x^* be a feasible solution of (P) which is convex program (f, g_i are convex functions, h_i are linear functions). If there exists (λ, μ) such that:
 - $\nabla f(x^*) + \sum_{i=1}^m \lambda_i \nabla g_i(x^*) + \sum_{i=1}^p \mu_i \nabla h_i(x^*) = 0$
 - $\lambda \geq 0$,
 - $\lambda_i g_i(x^*) = 0, i = 1, \dots, m$then x^* is a global optimal solution of (P)

KKT conditions - example

$$\begin{aligned} &\text{minimize } f(x,y) = 2x - y \\ &\text{s.t.} \quad g_1(x,y) = x^2 + y^2 - 2 \leq 0 \\ &\quad \quad g_2(x,y) = x - y - 1 \leq 0 \end{aligned}$$

- $\nabla f(x,y) = [2 \ 1]^T$, $\nabla g_1(x,y) = [2x \ 2y]^T$, $\nabla g_2(x,y) = [1 \ -1]^T$
- f and g_2 are linear, so they are convex
- $\nabla^2 g_1(x,y) = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ which is positive definite, so g_1 is convex

KKT conditions - example

minimize $f(x,y) = 2x - y$

s.t. $g_1(x,y) = x^2 + y^2 - 2 \leq 0$

$g_2(x,y) = x - y - 1 \leq 0$

Apply the KKT necessary conditions

$$[2 \ -1]^T + \lambda_1[2x \ 2y]^T + \lambda_2[1 \ -1]^T = 0$$

$$x^2 + y^2 - 2 \leq 0$$

$$x - y - 1 \leq 0$$

$$\lambda_1(x^2 + y^2 - 2) = 0$$

$$\lambda_2(x - y - 1) = 0$$

$$\lambda_1, \lambda_2 \geq 0$$

KKT conditions - example

$$\begin{aligned} &\text{minimize } f(x,y) = 2x - y \\ &\text{s.t.} \quad g_1(x,y) = x^2 + y^2 - 2 \leq 0 \\ &\quad \quad g_2(x,y) = x - y - 1 \leq 0 \end{aligned}$$

- Apply the KKT necessary conditions (rewrite)
 - $2 + 2\lambda_1 x + \lambda_2 = 0$ (1)
 - $-1 + 2\lambda_1 y - \lambda_2 = 0$ (2)
 - $x^2 + y^2 - 2 \leq 0$ (3)
 - $x - y - 1 \leq 0$ (4)
 - $\lambda_1(x^2 + y^2 - 2) = 0$ (5)
 - $\lambda_2(x - y - 1) = 0$ (6)
 - $\lambda_1, \lambda_2 \geq 0$ (7)

KKT conditions - example

$$\begin{aligned} &\text{minimize } f(x,y) = 2x - y \\ &\text{s.t.} \quad g_1(x,y) = x^2 + y^2 - 2 \leq 0 \\ &\quad \quad g_2(x,y) = x - y - 1 \leq 0 \end{aligned}$$

- There are 4 cases
 - (I) $\lambda_1 = \lambda_2 = 0$
 - (II) $\lambda_1 = 0, \lambda_2 > 0$
 - (III) $\lambda_1 > 0, \lambda_2 = 0$
 - (IV) $\lambda_1 > 0, \lambda_2 > 0$

KKT conditions - example

$$\begin{array}{ll}\text{minimize} & f(x,y) = 2x - y \\ \text{s.t.} & g_1(x,y) = x^2 + y^2 - 2 \leq 0 \\ & g_2(x,y) = x - y - 1 \leq 0\end{array}$$

- Case (I) $\lambda_1 = \lambda_2 = 0$, from (1) and (2) $\rightarrow$ NOT POSSIBLE

KKT conditions - example

minimize $f(x,y) = 2x - y$

s.t. $g_1(x,y) = x^2 + y^2 - 2 \leq 0$

$g_2(x,y) = x - y - 1 \leq 0$

- Case (II) $\lambda_1 = 0, \lambda_2 > 0$, from (1) and (2) we have
 - $2 + \lambda_2 = 0$
 - $-1 - \lambda_2 = 0$ (NOT POSSIBLE as $\lambda_2 > 0$)

KKT conditions - example

minimize $f(x,y) = 2x - y$

s.t. $g_1(x,y) = x^2 + y^2 - 2 \leq 0$

$g_2(x,y) = x - y - 1 \leq 0$

• Case (III) $\lambda_1 > 0, \lambda_2 = 0$, we have

• $2 + 2\lambda_1 x = 0$ (8)

• $-1 + 2\lambda_1 y = 0$ (9)

• $x^2 + y^2 - 2 = 0$ (10)

• $x - y - 1 \leq 0$ (11)

• Solve this equation, we get $x = -1$ and $y = 1$, $\lambda_1 = 1$, $\lambda_2 = 0$. This solution satisfies (1) - (7) $\rightarrow x = -1, y = 1$ is a global optimal solution to the problem (based on the KKT sufficient conditions)

KKT conditions - example

$$\begin{aligned} &\text{minimize } f(x,y) = 2x - y \\ &\text{s.t.} \quad g_1(x,y) = x^2 + y^2 - 2 \leq 0 \\ &\quad \quad g_2(x,y) = x - y - 1 \leq 0 \end{aligned}$$

- Case (IV) $\lambda_1 > 0, \lambda_2 > 0$, we have
 - $2 + 2\lambda_1 x + \lambda_2 = 0$ (12)
 - $-1 + 2\lambda_1 y - \lambda_2 = 0$ (13)
 - $x^2 + y^2 - 2 = 0$ (14)
 - $x - y - 1 = 0$ (15)
 - $\lambda_1, \lambda_2 \geq 0$ (16)

KKT conditions - example

$$\begin{aligned} &\text{minimize } f(x,y) = 2x - y \\ &\text{s.t.} \quad g_1(x,y) = x^2 + y^2 - 2 \leq 0 \\ &\quad \quad g_2(x,y) = x - y - 1 \leq 0 \end{aligned}$$

- Case (IV) $\lambda_1 > 0, \lambda_2 > 0$, we have
 - From (15), we have $x = y + 1$, replace in (12) and (13), we have
 - $2 + 2\lambda_1(y + 1) + \lambda_2 = 0$ (17)
 - $-1 + 2\lambda_1 y - \lambda_2 = 0$ (18)
 - Subtract (17) and (18), we have $3 + 2\lambda_1 + 2\lambda_2 = 0$ (NOT POSSIBLE as $\lambda_1 > 0, \lambda_2 > 0$)

KKT conditions - example

$$\begin{aligned} &\text{minimize } f(x,y) = 2x - y \\ &\text{s.t.} \quad g_1(x,y) = x^2 + y^2 - 2 \leq 0 \\ &\quad \quad g_2(x,y) = x - y - 1 \leq 0 \end{aligned}$$

- Case (IV) $\lambda_1 > 0, \lambda_2 > 0$, we have
 - From (15), we have $x = y + 1$, replace in (12) and (13), we have
 - $2 + 2\lambda_1(y + 1) + \lambda_2 = 0$ (17)
 - $-1 + 2\lambda_1 y - \lambda_2 = 0$ (18)
 - Subtract (17) and (18), we have $3 + 2\lambda_1 + 2\lambda_2 = 0$ (NOT POSSIBLE as $\lambda_1 > 0, \lambda_2 > 0$)



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THANK YOU !